

Assignment-02

Verification Planning Exercises

Exercise 1: 8-bit Adder

Prepare a complete verification plan for an 8-bit adder.

Identify:

- Features to be verified
 - Directed test cases
 - Corner cases
 - Functional coverage points
 - Scoreboard strategy
 - Assertions
 - Exit criteria
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Exercise 2: 8-bit Subtractor

Prepare a verification plan for an 8-bit subtractor.

Identify:

- Borrow generation scenarios
 - Underflow conditions
 - Directed tests
 - Corner cases
 - Coverage model
 - Scoreboard strategy
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Exercise 3: 4-bit Multiplier

Prepare a verification plan for a 4-bit multiplier.

Identify:

- Multiplication functionality
 - Maximum product conditions
 - Directed tests
 - Corner cases
 - Functional coverage
 - Reference model
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Exercise 4: 8-bit Comparator

Prepare a verification plan for an 8-bit comparator.

Verify:

- Equal condition
- Greater-than condition
- Less-than condition

Identify:

- Directed tests
 - Corner cases
 - Coverage points
 - Assertions
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Exercise 5: 4-to-1 Multiplexer

Prepare a verification plan for a 4-to-1 multiplexer.

Verify:

- All select line combinations
- Correct data routing

Identify:

- Directed tests
 - Corner cases
 - Coverage model
 - Assertions
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Exercise 6: 8-bit Encoder

Prepare a verification plan for an 8-bit encoder.

Identify:

- Valid encoding conditions
 - Invalid input combinations
 - Corner cases
 - Functional coverage points
 - Checker strategy
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Exercise 7: 3-to-8 Decoder

Prepare a verification plan for a decoder.

Identify:

- One-hot output conditions
 - Invalid conditions
 - Directed tests
 - Coverage requirements
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Exercise 8: 8-bit Priority Encoder

Prepare a verification plan for a priority encoder.

Identify:

- Priority handling
- Multiple active inputs
- Highest-priority selection

Prepare:

- Test scenarios
 - Corner cases
 - Coverage plan
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Exercise 9: 8-bit Up Counter

Prepare a verification plan for an up counter.

Verify:

- Reset operation
- Count increment operation
- Maximum count rollover

Identify:

- Corner cases
 - Coverage points
 - Assertions
-

Exercise 10: Up-Down Counter

Prepare a verification plan for an up-down counter.

Verify:

- Up counting
- Down counting
- Mode switching
- Overflow and underflow

Prepare:

- Directed tests
 - Corner cases
 - Coverage plan
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Exercise 11: Shift Register

Prepare a verification plan for a shift register.

Verify:

- Left shift
- Right shift
- Reset operation

Identify:

- Test scenarios
 - Corner cases
 - Functional coverage
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Exercise 12: Ring Counter

Prepare a verification plan for a ring counter.

Verify:

- One-hot operation
- Rotation sequence
- Reset functionality

Prepare:

- Assertions
 - Coverage model
 - Corner cases
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Exercise 13: Sequence Detector (1011)

Prepare a verification plan for a Mealy sequence detector.

Verify:

- Correct sequence detection
- Overlapping sequences
- Reset conditions

Prepare:

- Coverage plan
 - Assertions
 - Directed tests
-

Exercise 14: Traffic Light Controller

Prepare a verification plan for a traffic light controller.

Verify:

- State transitions
- Timing constraints
- Illegal state recovery

Prepare:

- Corner cases
 - Functional coverage
 - Assertions
-

Exercise 15: Synchronous FIFO

Prepare a verification plan for a FIFO.

Verify:

- Write operation
- Read operation
- Full condition
- Empty condition
- Simultaneous read/write

Prepare:

- Reference model
 - Coverage plan
 - Corner cases
 - Assertions
-

Exercise 16: ALU

Prepare a verification plan for an 8-bit ALU supporting:

- ADD
- SUB
- AND
- OR
- XOR
- NOT

Identify:

- Functional features
 - Directed tests
 - Corner cases
 - Coverage requirements
 - Scoreboard strategy
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Exercise 17: Register File

Prepare a verification plan for a register file.

Verify:

- Read operation
- Write operation
- Simultaneous read/write
- Reset functionality

Prepare:

- Coverage model
 - Assertions
 - Corner cases
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Exercise 18: Memory (RAM)

Prepare a verification plan for a RAM.

Verify:

- Read cycles
- Write cycles
- Consecutive accesses
- Boundary addresses

Prepare:

- Checker strategy
 - Coverage points
 - Corner cases
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Exercise 19: Arithmetic Logic Unit with Flags

Prepare a verification plan for an ALU that generates:

- Carry Flag
- Zero Flag
- Overflow Flag
- Sign Flag

Identify:

- Flag verification scenarios
 - Corner cases
 - Coverage requirements
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Exercise 20: UART Transmitter

Prepare a verification plan for a UART transmitter.

Verify:

- Start bit generation
- Data transmission
- Stop bit generation
- Baud-rate operation

Prepare:

- Functional coverage
- Assertions
- Scoreboard strategy
- Corner cases